Brandon Nguyen - Pathway 1: Sensing the World Reflection Due date: 09/23/2026

You are expected to type your reflection in essay format. The reflection should start with a personalized summary of what was done that week and how it was accomplished. Then you will describe what you learned and your experience. This should all be from your perspective, so pronouns are expected.

During the second week of labs, we went to Keysight to start working on our breadboards. On September 14th (first Keysight lab), we were given the task to build our first circuit with a solderless breadboard using a potentiometer as our sensor. Since I had prior experience using solderless breadboards compared to my teammates, who had no prior experience, I allowed them to do the first of many of the assigned circuit boards of this week. Serving as a helping hand, one partner copied the breadboard shown on the packet. Once we fully finished copying the first version of the breadboard, my other partner, who was closest to the power supply, hooked the breadboard up to channel 2 with the voltage set to 5. On the first attempt, if I remember correctly, it didn't work. So we asked for some help and moved some of the components around, which ultimately led to the 2nd attempt successfully working. After getting that checked off, we began working on the question, “What current is necessary to exceed .25W with 10 ohms of resistance?”In whichh we calculated using the formula sqrt(P/R)= I -> sqrt(.25/10) = .158 amps. After solving the question, we moved on to the next experiment where we put the pot to its maximum, ran 20 V, and then lowered the pot resistance all the way down. This experiment led to the resistor burning up and the LED no longer working at 5 V.

We then moved on to the switch-controlled diagram as one of my partners began copying the breadboard given to us in the package. After copying it, we ran it through the power source, and we successfully got it to work on the first try, with the switch button being able to turn the LED off. We then moved on to the environmental sensor as we swapped the switch sensor with a light sensor module, and we as a group began swapping out parts, using our intuition and the skills we’ve gathered so far to get the environmental sensor to work. We then answered the respective question assigned to the experiment before movingon too building a prototype of Dr. Valavanos system. After checking our BOM (bill of materials), the group decided on a secure and armed alarm system where the red LED would represent an armed intruder while the green LED represented a secure system with no intruders. The contact switch, the sensor we used in the experiment, served as a manual disarm or arm feature for the security system, where the default setting would be the green LED always being on; however, if it needed to be armed, pressing the switch would be capable of arming the system or disarming the system if the red LED was already on. One of my partners would then begin pinning the LED’s to the Arduino MicroPython while the other began journaling down which feature went to which pin. After assigning the pins, we would move on to the next step as we plugged in a default prompt into ChatGPT, with the result being nothing of value. We then constructed a prompt of our own which, summed up, told

ChatGPT that the green LED will be turned on by default as it's assigned to pin D3, and when the contact switch (connected to A3) is pressed, it would turn off the green LED and trigger the red LED (connected to A7)to flash on and off (like a blinker). After running the prompt through ChatGPT and pasting it into our Arduino, the system worked on the first try as intended. And so, since no changes were needed, we stated that in our packet and got it checked off. We would then update our repo as a group and clean the bench before leaving the September 14th lab.

Returning to the lab on September 16th, we would be assigned a new task of first programming a motor. As a group, we began going through questions like inputs, GPIO PINs, block diagram, etc. We would then move on to the physical construction, where pretty much everyone collaborated. One partner was working on the breadboard while my other partner and I gave our inputs and potential solutions. It was a bit challenging at first, but after asking for help and successfully applying the power, ground, and signal wires to our Arduino, we began constructing our ChatGPT prompt for our code. We asked ChatGPT to write us a MicroPython code for a breadboard connected to an Arduino Nano ESP32. How the system would work is that once the contact switch (connected to A6) was pressed, the micro servo 9g motor (connected to D5) would rotate 180 degrees. And when the switch is pressed again, the motor would reset back to the 0-degree angle it was originally at. After running our first trial, the code was unsuccessful, as it moved the motor without us pressing the contact switch. It also didn't reset to its original position after pressing the switch. After contemplating what went wrong, we realized the code was completely fine, but what most likely happened was a misplaced wire. So after placing the wire in the right place and running the experiment again, we got our intended outcome. We would then add header blocks to our code, upload it to our repo, and get that part checked off our packets. Moving on to the sensor performance data collection part of the experiment, we would begin reconstructing our breadboard to make it look like the diagram shown. We would then set the DC supply to 3.3 V and 1A as we set up the multimeter with clips for the experiment. When everything was done and ready, we would begin the experiment, as I was assigned to journal all of the data we've gathered; one partner focused on measuring the voltage, and the other partner focused on blowing on it, half-wipe with a wet wipe, wipe the entire surface, and hold the wipe on the entire surface. From this, we collected some pretty repetitive data, which we were told by an instructor was okay (we thought we had some sort of error, nd so we asked if what we've gathered was right). We would then answer the questions as we assigned our Arduino input as A7, and one of my partners reworked the breadboard as needed. We would then write a prompt for ChatGPT in which we basically asked it to convert the ADC readings into voltage and save the ADC values and readings into a CSV file, where it should include a time or sample number for each reading. We also emphasized the need for the CSV file to have heading columns, ADC value, and voltage, and that it should take new readings every second. After running it through our Arduino, the code successfully worked as intended,d with everything like recorded voltages and column headings being present throughout the experiment. After making the appropriate changes so that it gets 10 samples of data, we ran the experiment for all conditions and named the files after their condition for easier management. We would then put all of the data we

gathered onto a Google Sheet as we calculated averages and standard deviation. We also put a graph into Excel before uploading every file and Excel to our repo. We also decided to work on the questions at home since it was getting pretty late and people had things to do after the lab.

Overall, this week's lab was pretty time-consuming and pretty difficult compared to the materials engineering week of labs,s as the September 16th lab not only took the most time but also the most trial and error out of all of the labs so far. What I mainly learned from this experiment was how to use the Arduino MicroPython on a breadboard and how to use ChatGPT properly for code. And while I had prior experience with solderless breadboards, I believe the microcontroller experiment as a whole served as not only a great refresher to my prior experience but also taught me how to better wire, connect, and manage a breadboard when it comes to keeping it clean and compact. One thing I thought was really fascinating was the vast number of sensors that were actually available, as I never really played with a photoresistor light sensor before, and so that was a nice experience as I familiarized myself with how it worked. When it came to team communication, I believe that, due to the trial and error we experienced, our communication skills grew as a whole as we became more talkative and synergized way better compared to the material science week. So while the September 16th lab may have been a real pain, I believe it greatly helped us get familiar with ChatGPT, Arduino, and the breadboard while also improving our teamwork by vast amounts.